

SYED ALI SHAHREZ

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EDUCATION

Forman Christian College University, Lahore | BS Computer Science (Minor: Mathematics)

2023 – Present | Vice Rector's List (Fall 2023, Spring 2024) | Member, IEEE FCCU Chapter | Runner-Up, XR Hackathon 2.0 (2025)

Lahore Grammar School | A-Levels

2020 – 2022 | Mathematics, Computer Science, Physics, English Literature | Full Scholarship

Relevant Coursework: Artificial Intelligence, Design & Analysis of Algorithms, Data Structures & Algorithms, Parallel & Distributed Computing, Compiler Construction, Computer Networks, Object-Oriented Programming, Digital Logic Design.

Upcoming: Machine Learning, Deep Learning (elective coursework).

EXPERIENCE

FlyRank | Remote

Machine Learning Intern | July 2026 – Present

- Trained and diagnosed a Random Forest decline-risk ranking model on a client-holdout split; a perfect training score initially read as overfitting, but a regularization sweep failed to behave the way that diagnosis predicted, correctly redirecting to the real cause — a 618-row test set with 96.8% of rows from a single client.
- Traced a separate ranking-formula underperformance to a feature-scale mismatch and fixed it by switching to percentile ranks before combining features, raising Precision@20 from 0.450 to 0.800.

AI Fluency Intern | July 2026 – Present

- Built AI agents and automated workflows using the Model Context Protocol (MCP), including MCP servers, agent skills, subagents, and tool integrations.
- Applied Claude API concepts — RAG, context window management, Projects, and Connectors — in hands-on exercises as part of a technical AI-fluency training track.

PROJECTS

Lung Cancer Detection — Ensemble Deep Learning | Python, TensorFlow, Keras, CNNs

github.com/Ali-Shahrez/Lung-Cancer-AI-Ensemble-CNN

- Built an ensemble of 3 convolutional neural networks for binary classification of lung CT scans (malignant vs. normal/benign), trained on 1,097 images from the IQ-OTHNCCD dataset, achieving 80.82% accuracy.
- Designed a 5-block CNN architecture with batch normalization and L2 regularization; built a Tkinter GUI for real-time inference with confidence scores.

OrderBot — Conversational AI Ordering Assistant | Python, Streamlit, Ollama (LLM)

github.com/Ali-Shahrez/orderbot-ai

- Built an LLM-powered chatbot for natural-language food ordering that parses conversational input into structured order data and computes billing.
- Ran fully locally via Ollama, removing dependency on a paid API for language model inference.

Distributed MOT Vehicle Database Analytics | Python, mpi4py (MPI), Tkinter, matplotlib

github.com/Ali-Shahrez/Parallel_Project

- Built a distributed data analytics application over a ~3.5GB UK government vehicle-testing dataset, sharding data across MPI worker processes for parallel search and analysis.
- Implemented distributed search/filtering and MPI-reduced pass-rate analysis with matplotlib visualization; added persistent shard caching for ~10–50x faster warm starts.
- Deployed and verified on a genuine two-machine cluster over WiFi (MS-MPI); covered by 131 automated test groups across 8 development phases. COMP 410 (Parallel and Distributed Computing) semester project.

SKILLS

Languages: Python, HTML, CSS, SQL

AI/ML: TensorFlow, Keras, Convolutional Neural Networks (CNNs), ensemble modeling, deep learning, LLM integration (Ollama)

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Digital Logic Design, relational database design, complex SQL queries, data normalization

Tools: Git, VS Code, SQL Server, Microsoft Office

AWARDS

- High Achiever's Award (2020)
- Runner-Up, XR Hackathon 2.0 (2025)

Languages: English, Urdu, Punjabi, German (A1)